Review Test Submission: Assignment Week 9

User	Mr W (Wouter) W Nel
Course	EBN 122 S2 2017
Test	Assignment Week 9
Started	9/18/17 8:18 AM
Submitted	9/25/17 1:52 PM
Due Date	9/25/17 11:59 PM
Status	Completed
Attempt Score	24.2 out of 30 points
Time Elapsed	173 hours, 33 minutes

Instructions

Read over the document *ClickUP Assignments Instructions*.

The deadline is the time by which the assignment must be SUBMITTED (not just started)

Before sending an enquiring regarding ClickUP question to any-one please verify that your problem is valid ... i.e. if you suspect incorrect marking first check with a fellow student or two.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final correct answer 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- Use SI units unless otherwise specified.
- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use sigular)
- Use '**u**' to indicate micro.
- Only use pico(**p**), nano(**n**), micro(**u**), milli(**m**), kilo(**k**), mega(**M**), giga(**G**) and tera(**T**); do not use centi, deci and deca

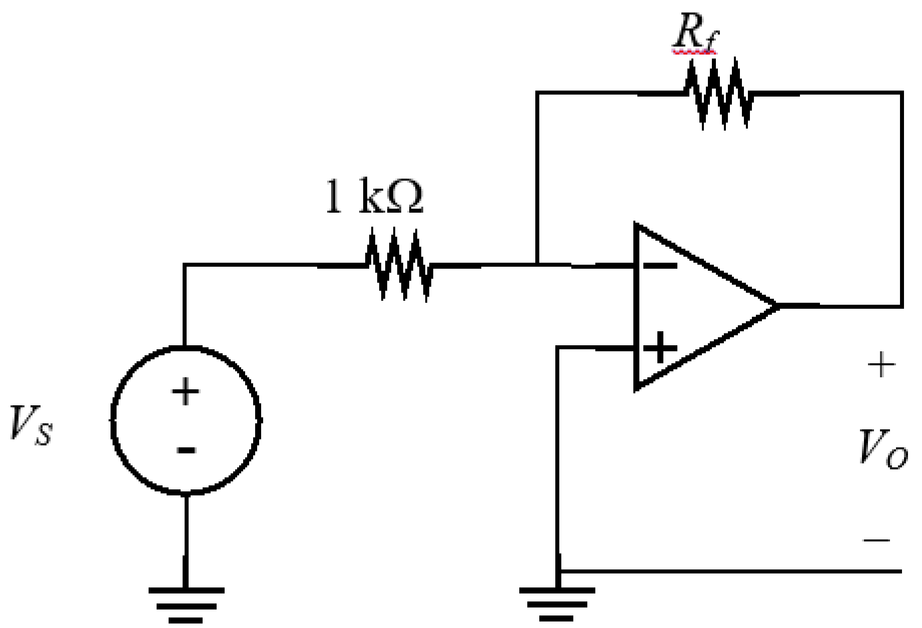
It does not have meaning to use milli(m) or kilo(k) when not as part of a unit. So when no units are required just write 0.1 or 1000.

Results Displayed	All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions

Question 1

0.2 out of 2 points

The range of the output voltage V_O is: $-10 \leq V0 \leq -1$. The range of the input voltage Vs is: $1 \leq Vs \leq 9.9$. Determine the maximum value of R_f (kOhm) which will satisfy the above constraints.



Selected Answer: 9.9kohm

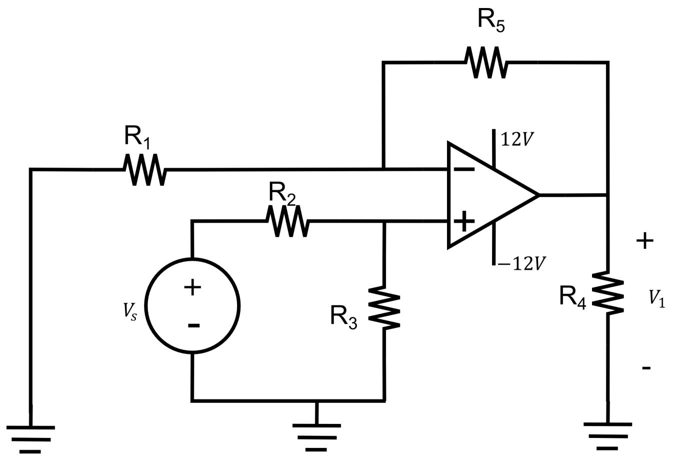
Correct Answer: 1.01 ± 5% (kohm)

Question 2

2 out of 2 points

What is the maximum value V_S can assume to ensure that the operational amplifier operates in its linear range?

R_1=5Ω, R_2=34Ω, R_3=11Ω, R_4=5Ω and R_5=26Ω



Selected Answer: 7.92V

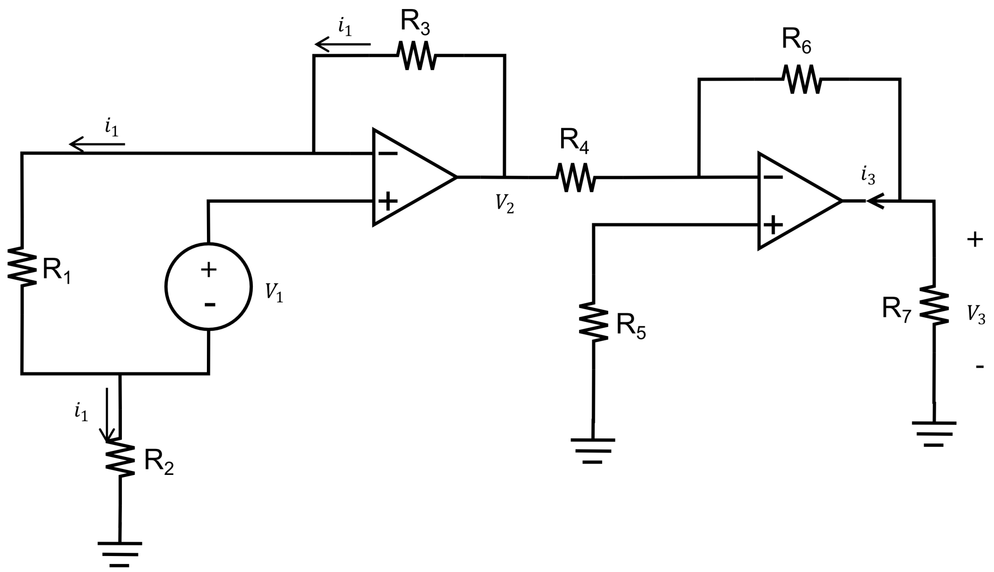
Correct Answer: 7.918 ± 2% (V)

Question 3

2 out of 2 points

Determine the amplification factor V3/V_2

R_1=8kΩ, R_2=5kΩ, R_3=17kΩ, R_4=4kΩ, R_5=5kΩ, R_6=16kΩ and R_7=6kΩ



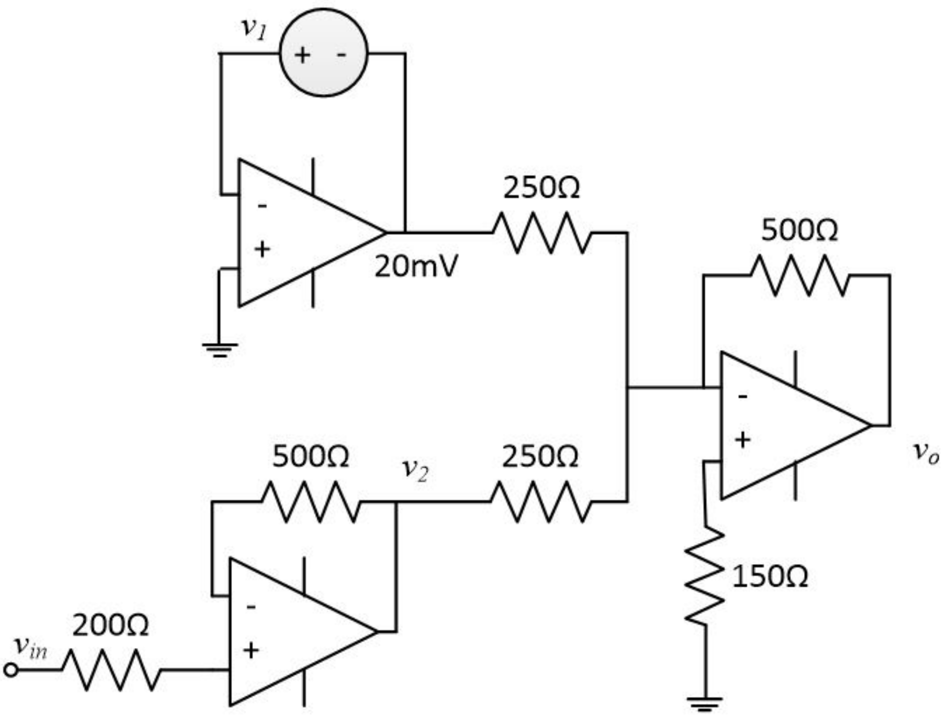
Selected Answer: -4

Correct Answer: -4.000 ± 2%

Question 4

2 out of 2 points

Find vo if Vin = 7 V



Selected Answer: -14.04V

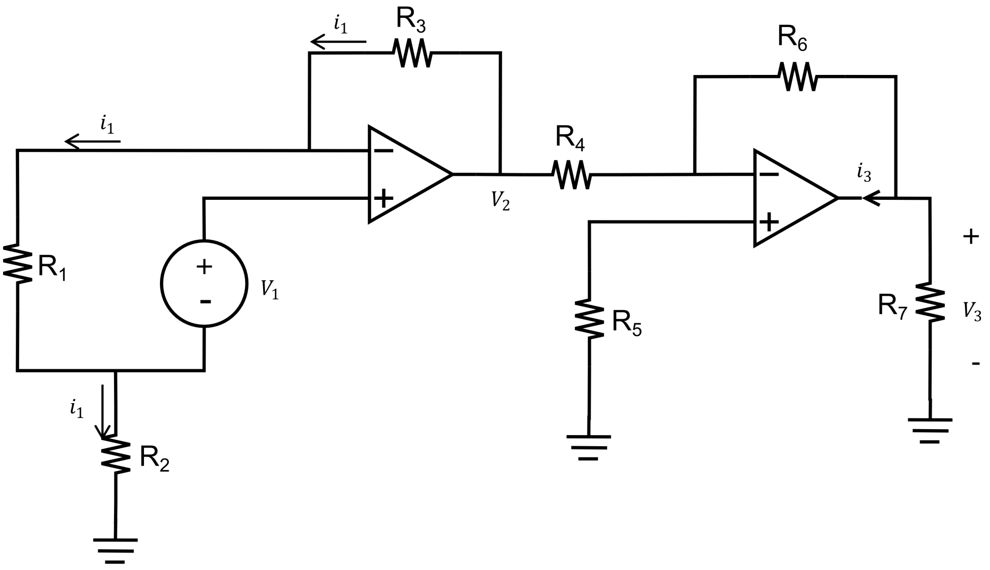
Correct Answer: -14.04 ± 5% (V)

Question 5

0 out of 2 points

Determine the amplification factor V_2/V_1

R_1=6kΩ, R_2=7kΩ, R_3=14kΩ, R_4=5kΩ, R_5=5kΩ, R_6=10kΩ and R_7=2kΩ



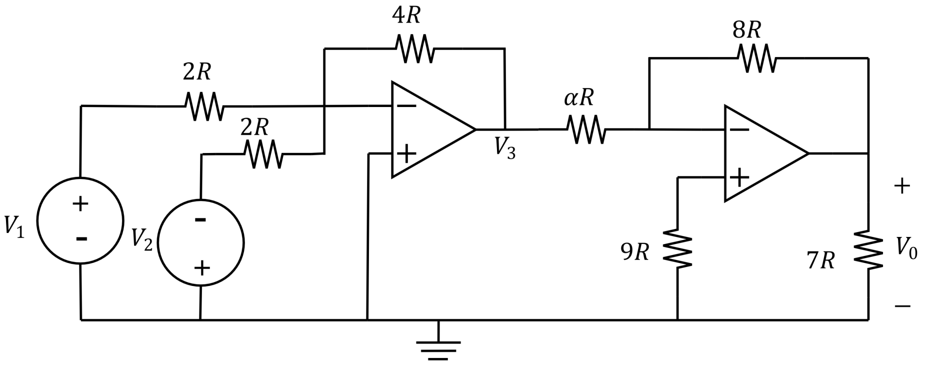
Selected Answer: 0.19

Correct Answer: 4.500 ± 2%

Question 6

2 out of 2 points

If V_3=9V and V_2=9V determine the value for α which will result in V_0=-14V



Selected Answer: 5.14

Correct Answer: 5.143 ± 2%

Question 7

0.2 out of 2 points

The voltage across a 9μF capacitor is given as $2+8e^{-21t}$. Determine the current at t=40ms

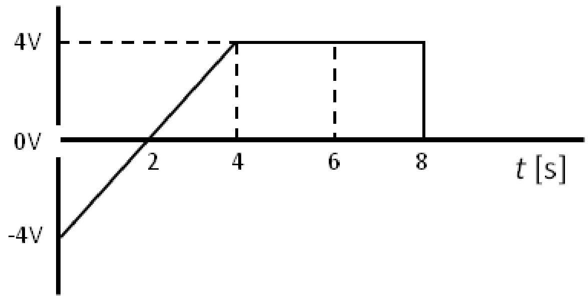
Selected Answer: 49.1uA

Correct Answer: -652.746 ± 2% (uA)

Question 8

2 out of 2 points

The graph below shows the voltage across a 3H inductor. Assuming the initial current through the inductor is 9 A, what is the current at t=5.7 s?



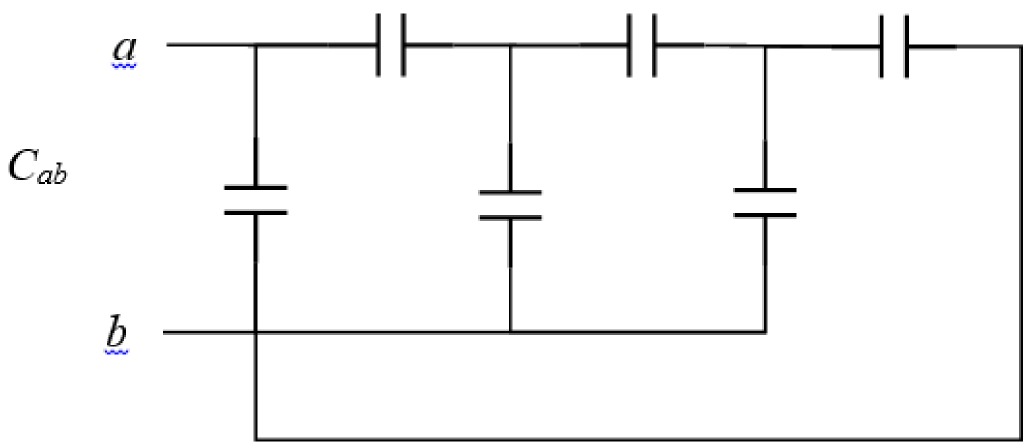
Selected Answer: 11.3A

Correct Answer: 11.267 ± 2% (A)

Question 9

1.8 out of 2 points

Determine the equivalent capacitance Cab of the circuit. Every capacitor has a value of 13 mF. Give your answer in mF as well.



Selected Answer: 21.1uF

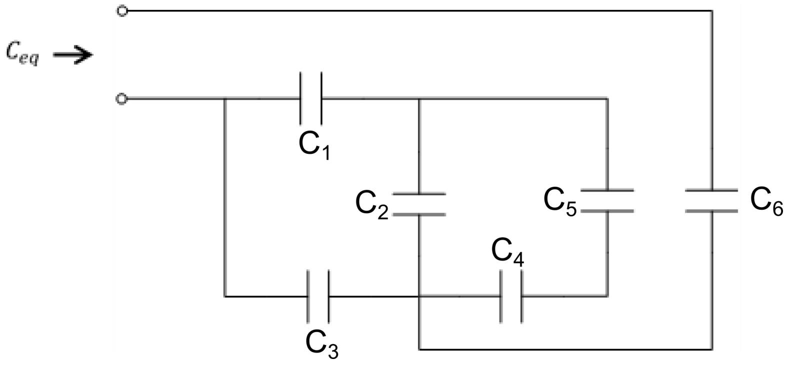
Correct Answer: 21.13 ± 5% (mF)

Question 10

2 out of 2 points

Determine C_{eq}

C_1=537nF, C_2=539nF, C_3=443nF, C_4=539nF, C_5=539nF and C_6=378nF



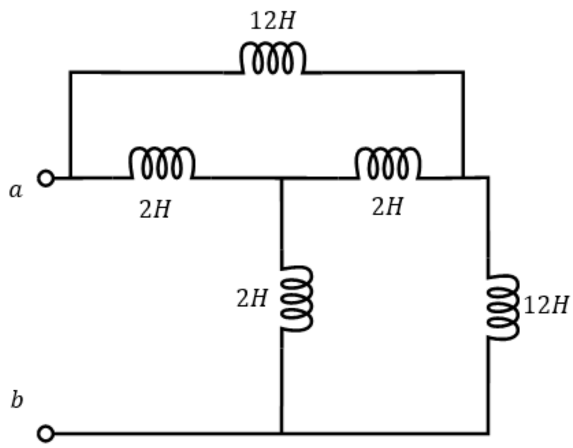
Selected Answer: 253nF

Correct Answer: 253.066 ± 2% (nF)

Question 11

2 out of 2 points

Find the equivalent inductance L_{ab}.



Selected Answer: 3.43H

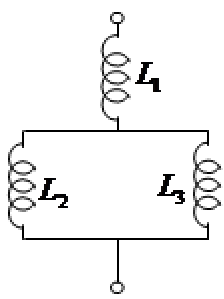
Correct Answer: 3.429 ± 2% (H)

Question 12

2 out of 2 points

Find the equivalent inductance to the combination below.

L_1=5mH, L_2=12mH and L_3=11mH,



Selected Answer: 10.7mH

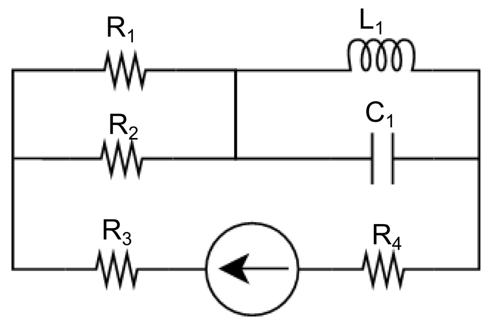
Correct Answer: 10.739 ± 2% (mH)

Question 13

2 out of 2 points

Calculate the energy stored by the inductor (w_L) Assume steady state (direct current) conditions.

R_1=98kΩ, R_2=56kΩ, R_3=63kΩ, R_4=96kΩ, C_1=64nF, L_1=19mH and the current source, I_S =88mA.



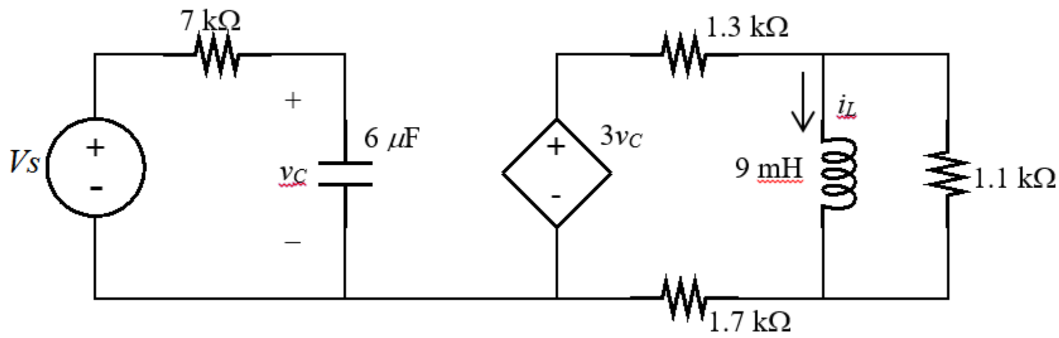
Selected Answer: 73.6uJ

Correct Answer: 73.568 ± 2% (uJ)

Question 14

2 out of 2 points

Determine the energy stored in the inductor, given Vs = 26.



Selected Answer: 3.04uJ

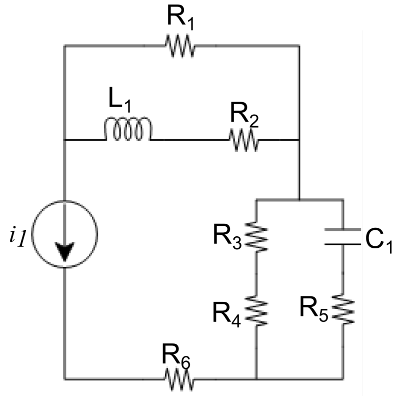
Correct Answer: 3.04 ± 1% (uJ)

Question 15

2 out of 2 points

Calculate the energy stored by the capacitor (w_C). Assume direct current/steady state conditions.

R_1=7kΩ, R_2=7kΩ, R_3=46kΩ, R_4=43kΩ, R_5=7kΩ, R_6=2kΩ, C_1=6nF, L_1=1mH and i_1 =49mA



Selected Answer: 57.1mJ

Correct Answer: 57.055 ± 2 (mJ)

Monday, September 25, 2017 1:52:47 PM SAST

← **OK**